

 **Unsupported release:** Isaac Sim 5.1.0 is no longer supported. Bug fixes and new features are delivered only in newer releases.

Tutorial 6: Setup a Manipulator

Learning Objectives

This is the first manipulator tutorial in a series of four tutorials. This tutorial shows how to import the UR10e robot from Universal Robots and the 2F-140 gripper from Robotiq into NVIDIA Isaac Sim from URDF files and connect them together under one articulation.

30 Minutes Tutorial

Prerequisites

- If you are new to NVIDIA Isaac Sim, complete the [Wheeled Robot Set Up Tutorials](#) tutorial prior to beginning this tutorial.
- Review the ROS 2 installations [ROS 2 Installation](#) prior to beginning this tutorial.
- Review the URDF importer [URDF Importer Extension](#) tutorial.
- In a ROS sourced terminal, install xacro using the following command (Linux only):

```
sudo apt install ros-$ROS_DISTRO-xacro
```

- Locate the `import_manipulator` folder in the content browser at `Isaac Sim/Samples/Rigging/Manipulator/import_manipulator/`.

Note

The ROS URDF import steps are tested on Linux only, it may not work on Windows. If you are using Windows, you can skip the ROS import steps and use the USD files provided in the content browser.

Build and Install the UR Description Package (Linux only)

Isaac Sim 5.1 requires Python 3.11, which is not natively supported by the ROS 2 UR description package, so we need to build the package from source.

Note

See [Enabling rclpy, Custom ROS 2 Packages, and Workspaces with Python 3.11](#) for more information on setting up your custom ROS 2 package in your ROS workspace.

Clone the UR Description Package

1. Clone the UR description package from the [Universal Robots ROS 2 Description repository](#).

```
git clone https://github.com/UniversalRobots/Universal_Robots_ROS2_Descripti
```

2. Switch to the branch that matches your ROS distribution.

ROS 2 Humble

ROS 2 Jazzy

```
git checkout humble
```

3. Copy the repository into your Isaac Sim ROS Workspace `src` folder.

Build the UR Description Package Using Python 3.11

1. Go to the Isaac Sim ROS Workspace, and run the following command to build the UR description package using Python 3.11.

```
./build_ros.sh
```

2. Source the Python 3.11 ROS environment and launch Isaac Sim. Replace `<ROS distro>` with your ROS distribution (for example, `humble` or `jazzy`).

```
source build_ws/<ROS distro>/<ROS distro>_ws/install/local_setup.bash
source build_ws/<ROS distro>/isaac_sim_ros_ws/install/local_setup.bash
./path/to/isaac-sim.sh
```

Build the UR Description Package Using System ROS

1. Source your system ROS environment. Refer to [Setting Up Workspaces](#) for more information on setting up your ROS 2 workspace.
2. Navigate to your Isaac Sim ROS Workspace and run the following commands to build it:

```
rosdep install -i --from-path src --rosdistro $ROS_DISTRO -y  
colcon build  
source install/setup.sh
```

Import the UR10e Robot (Linux only)

Enable the ROS 2 Robot Description URDF Importer Extension

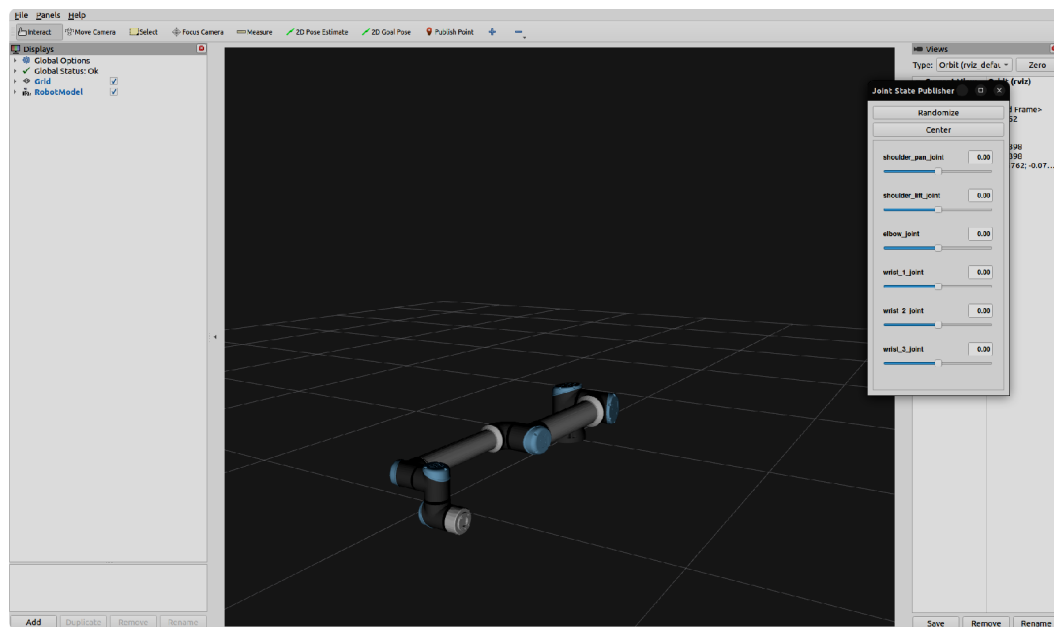
1. Go to **Window** > **Extensions**.
2. Type **URDF** in the search box, and find the **ROS 2 Robot Description URDF Importer Extension**.
3. If you can't find it, remove the **@feature** filter from the search box.
4. If you still can't find it, make sure Isaac Sim was launched from the same terminal where ROS was sourced.
5. Enable the extension by clicking the toggle button labeled **ENABLE**.
6. Check the box for **AUTOLOAD**, just to the right of **ENABLE**.

Launch the URDF Publisher Topic

1. In the system ROS sourced terminal that you created earlier, launch the UR10e description by running:

```
ros2 launch ur_description view_ur.launch.py ur_type:=ur10e
```

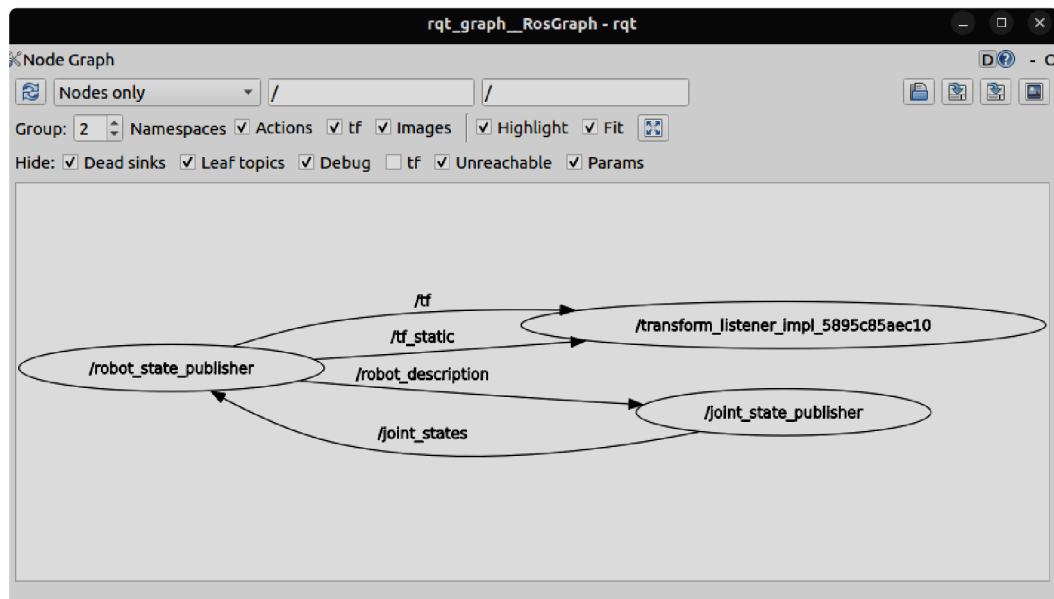
2. Verify that you see a window similar to the image below:



3. Set up one more terminal for `rqt_graph`, to see ROS nodes and topics being published:

```
rqt_graph
```

4. Verify that you see a window similar to the image below:



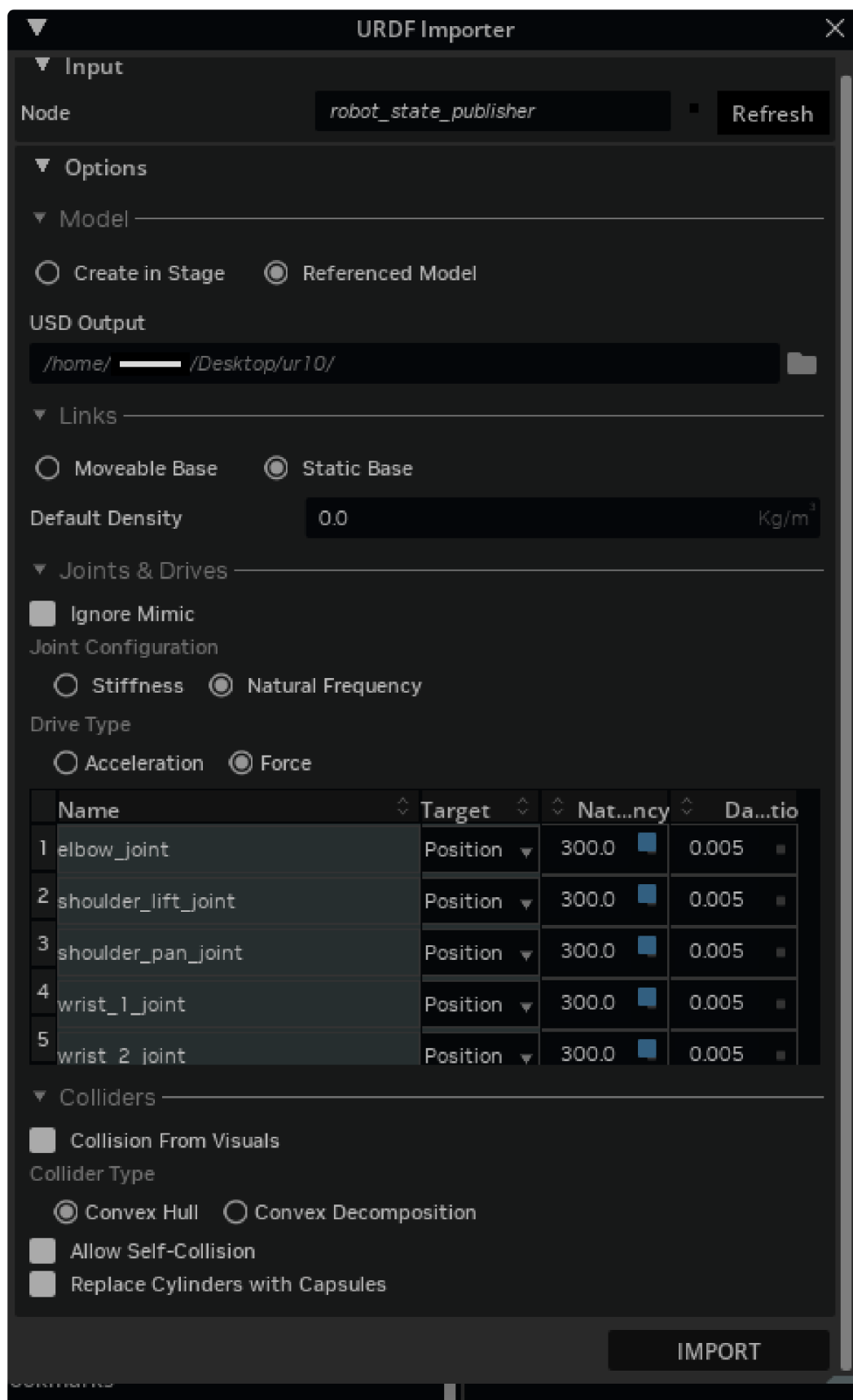
Hint

If the nodes are not showing up in `rqt_graph`, press the refresh button next to the drop down menu.

Import the UR10e Robot into Isaac Sim

1. Go to Isaac Sim.
2. Navigate to **File > Import from the ROS 2 URDF Node**.

- In the **Node** field, type `robot_state_publisher`, click **Refresh**.
- In the **Model** field, select the desired output (for example, `~/Desktop`).
- Select **Natural Frequency** for joint configuration.
- Select all the joints listed below, then set the **Natural Frequency** to `300` to ensure the joints are sufficiently stiff.



3. Click **Import**.

For reference, the resulting USD file is available in the content browser at [Isaac Sim/Samples/Rigging/Manipulator/import_manipulator/ur10e/ur/ur.usd](#).

Import the Robotiq 2F-140 Gripper (Linux only)

Use the URDF file provided by [ros-industrial-attic](#). Even though this package is built for ROS1 and is deprecated, you can still adopt the URDF files and import the gripper for NVIDIA Isaac Sim.

Convert XACRO to URDF

1. Download the Repository from [here](#).

```
git clone https://github.com/ros-industrial-attic/robotiq.git
```

2. Navigate to the `robotiq/robotiq_2f_140_gripper_visualization/urdf` folder, open each xacro file.

- Replace `$(find robotiq_2f_140_gripper_visualization)` with the relative path to the target file (for example, `robotiq_arg2f_transmission.xacro`) from the current xacro file.

For example, in `robotiq_arg2f_140_model.xacro`, replace:

```
<xacro:include filename="$(find robotiq_2f_140_gripper_visualization
```

With:

```
<xacro:include filename="../robotiq_arg2f_transmission.xacro" />
```

- Replace `package://` with the relative path to the target file (for example, `robotiq_arg2f_${stroke}_inner_finger.stl`) from the current xacro file.

For example, in `robotiq_arg2f_140_model.xacro`, replace:

```
<mesh filename="package://robotiq_2f_140_gripper_visualization/r
```

With:

```
<mesh filename="../meshes/visual/robotiq_arg2f_${stroke}_inner_finge
```

3. Convert the xacro files to URDF format:

```
xacro robotiq_arg2f_140_model.xacro > robotiq_2f_140.urdf
```

If you encounter the error `xacro: command not found`, you need to install xacro.

- Install xacro using the following command:

```
sudo apt install ros-$ROS_DISTRO-xacro
```

For reference, the resulting URDF files is available in the content browser at [Isaac Sim/Samples/Rigging/Manipulator/import_manipulator/robotiq_2f_140_urdf/urdf/roboti](#)

Import Robotiq 2F-140 Gripper into Isaac Sim

1. Go to Isaac Sim.
2. Let's create a new stage by going to **File > New**.
3. Navigate to **File > Import**.
4. Select the `robotiq_2f_140.urdf` file that you imported from the previous step.
5. In the import settings:
 - For USD Output, navigate to your desktop using file browser and select **Desktop** this will be the output location of the gripper USD.
 - For `finger_joint`, set the Natural Frequency to `300`.

- For the other joints of target **Mimic**, set the Natural Frequency to **2500**.

6. Click **Import** to complete the process.

Date Modified 01/29/2025 02:23PM
Created by
Modified by
File size 12.27 KB

▼ Options

▼ Model

☐ Create in Stage ☒ Referenced Model

USD Output
Same as Imported Model(Default)

▼ Links

☐ Moveable Base ☒ Static Base

Default Density 0.0 Kg/m³

▼ Joints & Drives

☐ Ignore Mimic

Joint Configuration
☐ Stiffness ☒ Natural Frequency

Drive Type
☐ Acceleration ☒ Force

	Name	Target	Nat...ncy	Da...tio
1	finger_joint	Position	300.0	0.005
2	left_inner_finger_joint	Mimic	2500.0	0.005
3	left_inner_knuckle_joint	Mimic	2500.0	0.005
4	right_inner_finger_joint	Mimic	2500.0	0.005
5	right_inner_knuckle_joint	Mimic	2500.0	0.005

▼ Colliders

☐ Collision From Visuals

Collider Type
☒ Convex Hull ☐ Convex Decomposition

☐ Allow Self-Collision

☐ Replace Cylinders with Capsules

For reference, the resulting USD file is available in the content browser at [Isaac Sim/Samples/Rigging/Manipulator/import_manipulator/robotiq_2f_140/robotiq_2f_140.usd](#).

Expected Parameters for Finger and Knuckle Joints

Joint Name	Lower Limit	Upper Limit	Gearing	Stiffness	Damping	Max Force
Finger Joint	0	40.107	N/A	37.51957	0.00125	1000
Left inner Finger	-8.021	48.128	-1	N/A	N/A	N/A
Left Inner Knuckle	-48.128	8.021	1	N/A	N/A	N/A
Right inner Knuckle	-48.128	8.021	1	N/A	N/A	N/A
Right outer knuckle	-48.128	8.021	1	N/A	N/A	N/A
Right inner Finger	-8.021	48.128	-1	N/A	N/A	N/A

Expected Parameters for Mimic Joints

- Reference Joint: `/robotiq_arg2f_140_model/joints/finger_joint`
- Reference Joint Axis: `rotX`
- Natural Frequency: `2500`
- Damping Ratio: `0.005`

Connect the UR10e Robot with the Robotiq 2F-140 Gripper

Much like a real robot can have its tools changed for different tasks, simulated robots benefit from the same capability. This section outlines two methods to connect the UR10e robot with the Robotiq 2F-140 gripper:

- Option 1**, shows how to connect the gripper to the robot directly using a fixed joint with a shared articulation.
- Option 2**, shows how to use the robot assembler and variant to connect the end effectors to the robot. Depending on the variant selected, the gripper will be added as a payload, which allows us to load or unload the different end effectors depending on which variant is enabled.

Option 1: Connect the UR10e with the Robotiq 2F-140 Gripper using the GUI

1. Open the UR10e USD file created from the last activity (`ur.usd`).
2. Drag and drop the `robotiq_2f_140.usd` file, we created earlier, into the stage.
3. Rename the `robotiq_2f_140.usd` prim to `ee_link`.
4. Set the `ee_link` xform to the position and orientation of `wrist_3_link`.

```
Translate (1.18425, 0.2907, 0.06085)
Orient (-90, 0, -90)
```

5. Select `ee_link/root_joint`.
6. Go to the `Physics Articulation Root` section in the Property Editor, remove the `Articulation Root`.

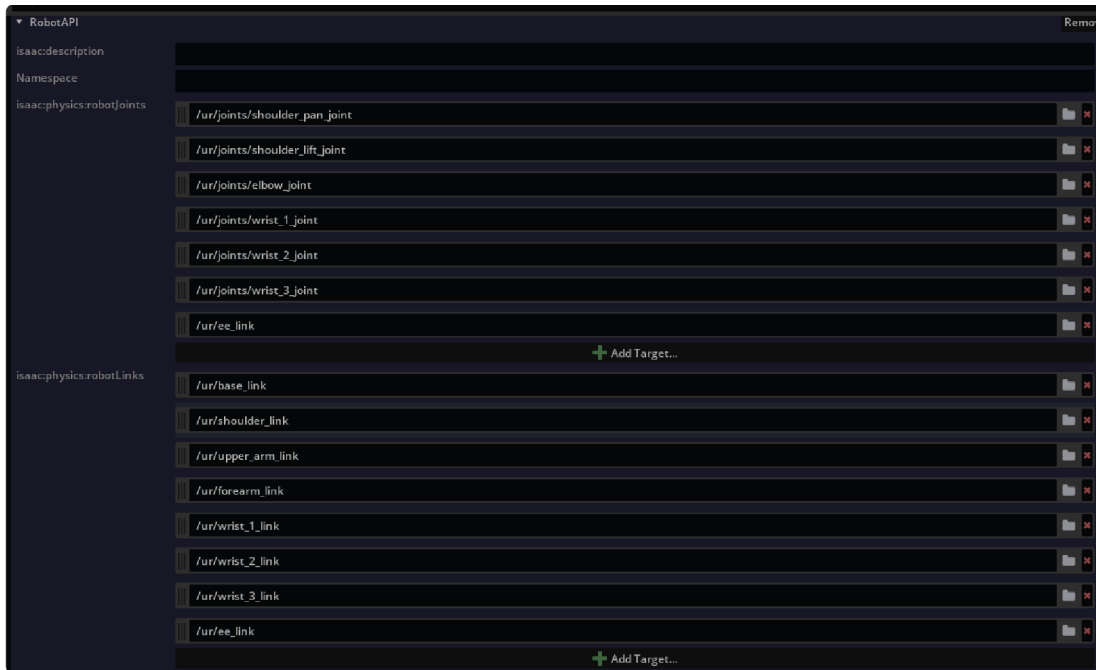
Only select a single articulation for the robot.

7. Go down to the `Joints` section in the Property Editor.
8. Set `Body0` to `/ur/wrist_3_link`, to joint the end effector to the robot.



Nest the UR10e robot schema into the 2F-140 gripper's robot schema:

1. Select the `ur` prim.
2. Go down to the `IsaacRobotAPI` section in the Property Editor, and add `/ur/ee_link` to both the `isaac:physics:robotjoints` and `isaac:physics:robotLinks` fields, to make sure that the UR10e robot's robot schema includes the 2F-140 gripper's robot schema.



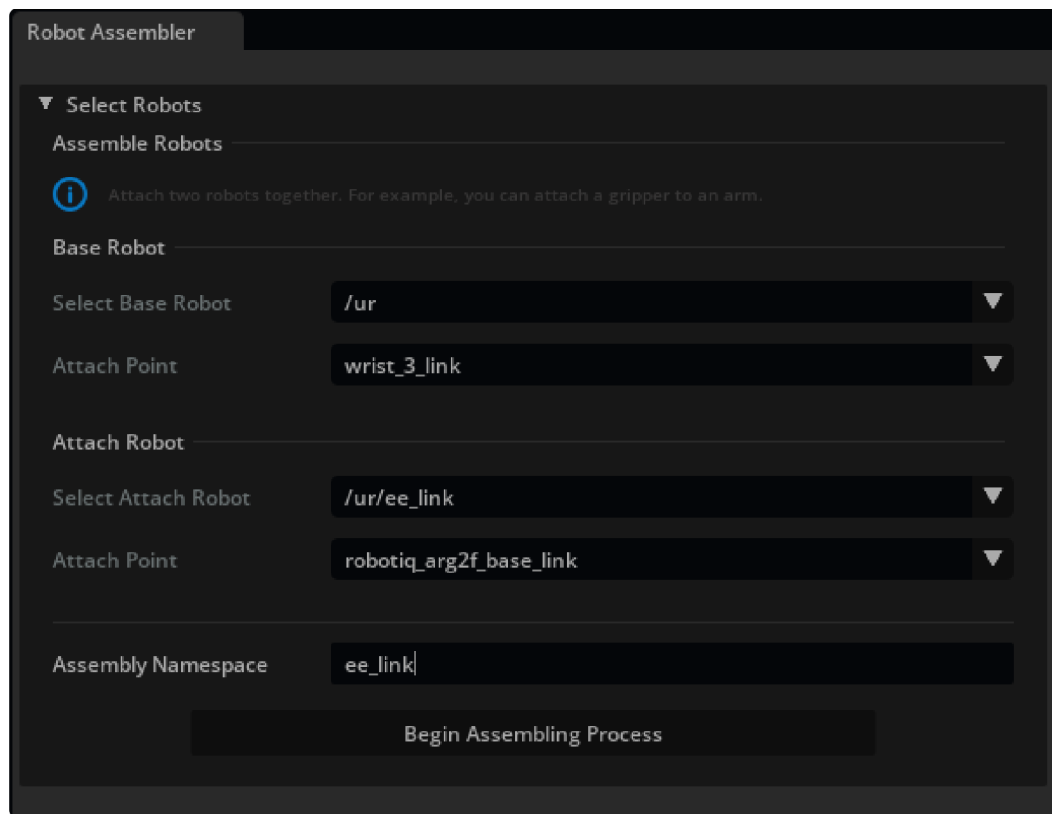
Your robot is now connected to the gripper, and you can test your robot in [Tutorial 6: Setup a Manipulator](#).

For reference, we also provide the resulting USD file in Content Browser at [Isaac Sim/Samples/Rigging/Manipulator/import_manipulator/ur10e/ur/ur_gripper_manual.usd](#).

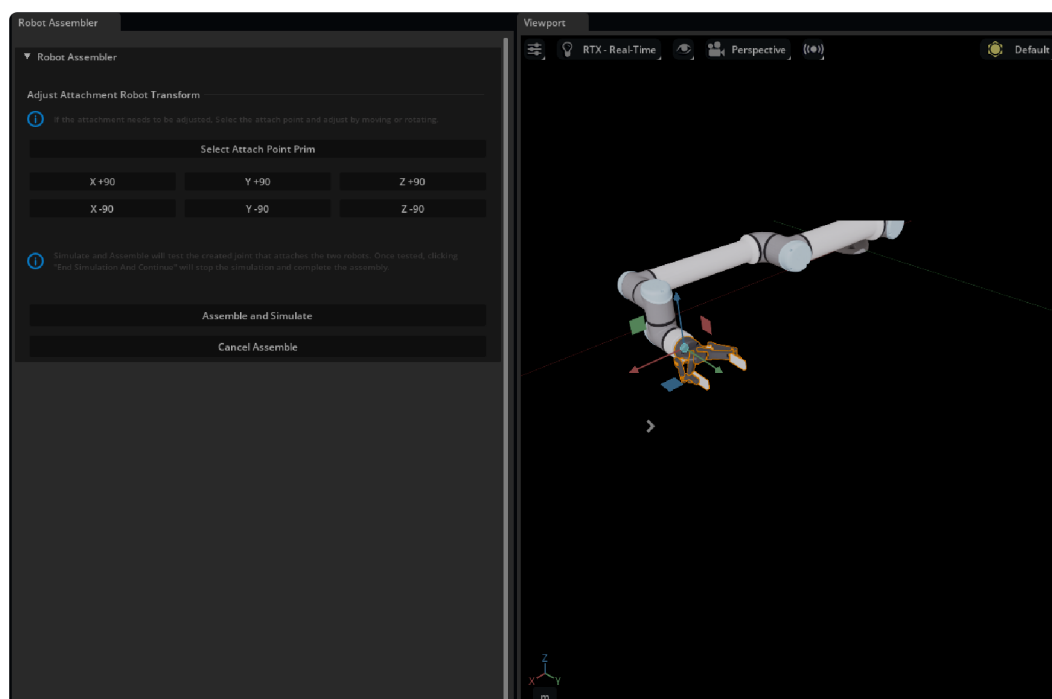
Option 2: Connect the UR10e with the Robotiq 2F-140 Gripper using the Robot Assembler

Alternatively, you can use the Robot Assembler to connect the UR10e with the Robotiq 2F-140 gripper. The robot assembler will add the gripper as a variant to a sublayer of the base robot, giving you greater flexibility to switch between different end effectors.

1. Open the UR10e USD file created from the last activity (`ur.usd`).
2. Drag and drop the `robotiq_2f_140.usd` file we created earlier into the stage.
3. Rename the `robotiq_2f_140` prim to `ee_link`.
4. Open the robot assembler by going to **Tools > Robotics > Asset Editor > Robot Assembler**.
 - In **Base Robot**, set **Select Base Robot** to `/ur`, **Attach Point** to `wrist_3_link`.
 - In **Attach Robot**, set **Select Attach Robot** to `/ur/ee_link`, **Attach Point** to `robotiq_arg2f_base_link`.
 - Set **Assembly Namespace** to `ee_link`.
5. Click **Begin Assembling Process** to start the process.



6. Adjust the attachment point orientation to make sure the end effector is attached to the gripper correctly. Rotate the gripper 90 degrees around the z-axis by clicking **Z +90**.

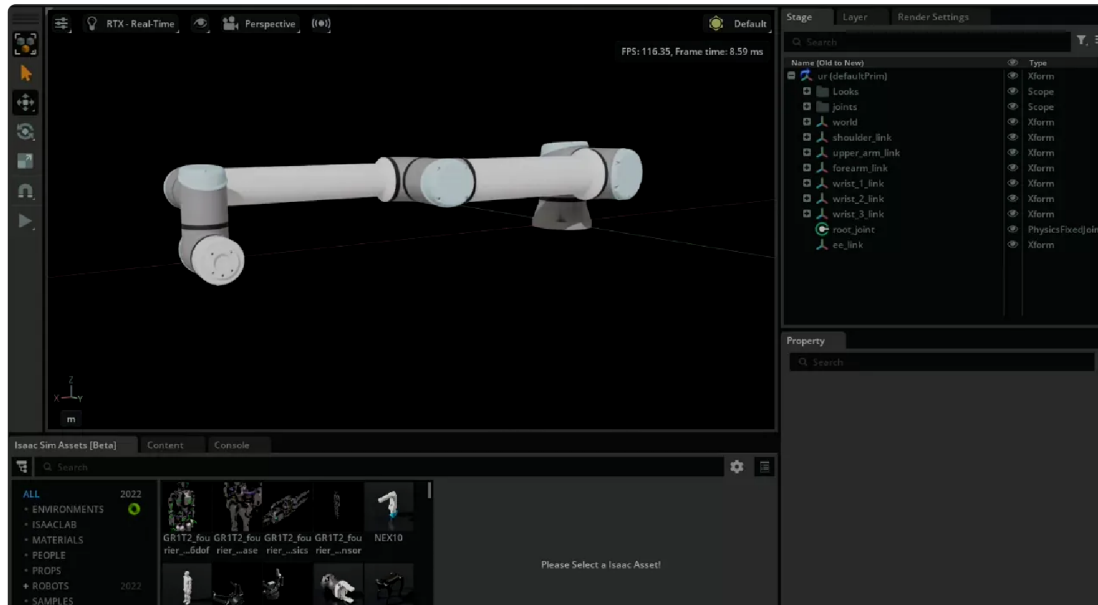


7. Click **Assemble and Simulate** to test the process.
8. Click **End Simulation And Finish** to complete the process.

Run the Simulation

1. In the Stage panel, select the **ur** prim.
2. In the Property Editor at the bottom right, find the **Variants** section.
3. Beside **ee_link**, select **None** and the gripper will be removed from the robot.

4. Beside **ee_link**, select **robotiq_2f_140** and the gripper will be added to the robot.
5. Save the asset by going to **File > Save** or press **Ctrl+S**.



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